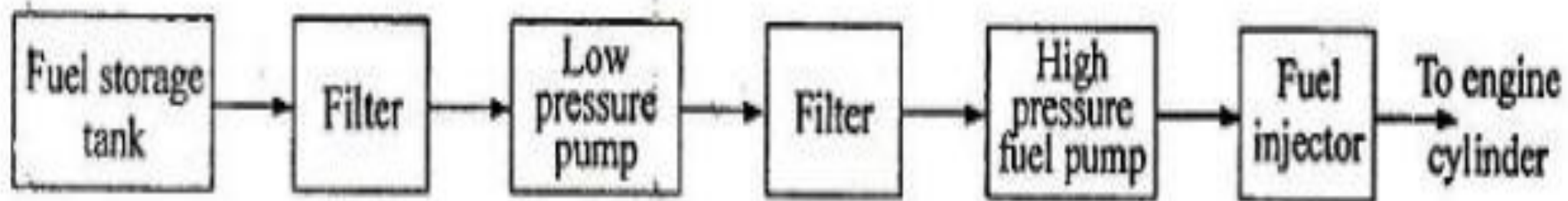


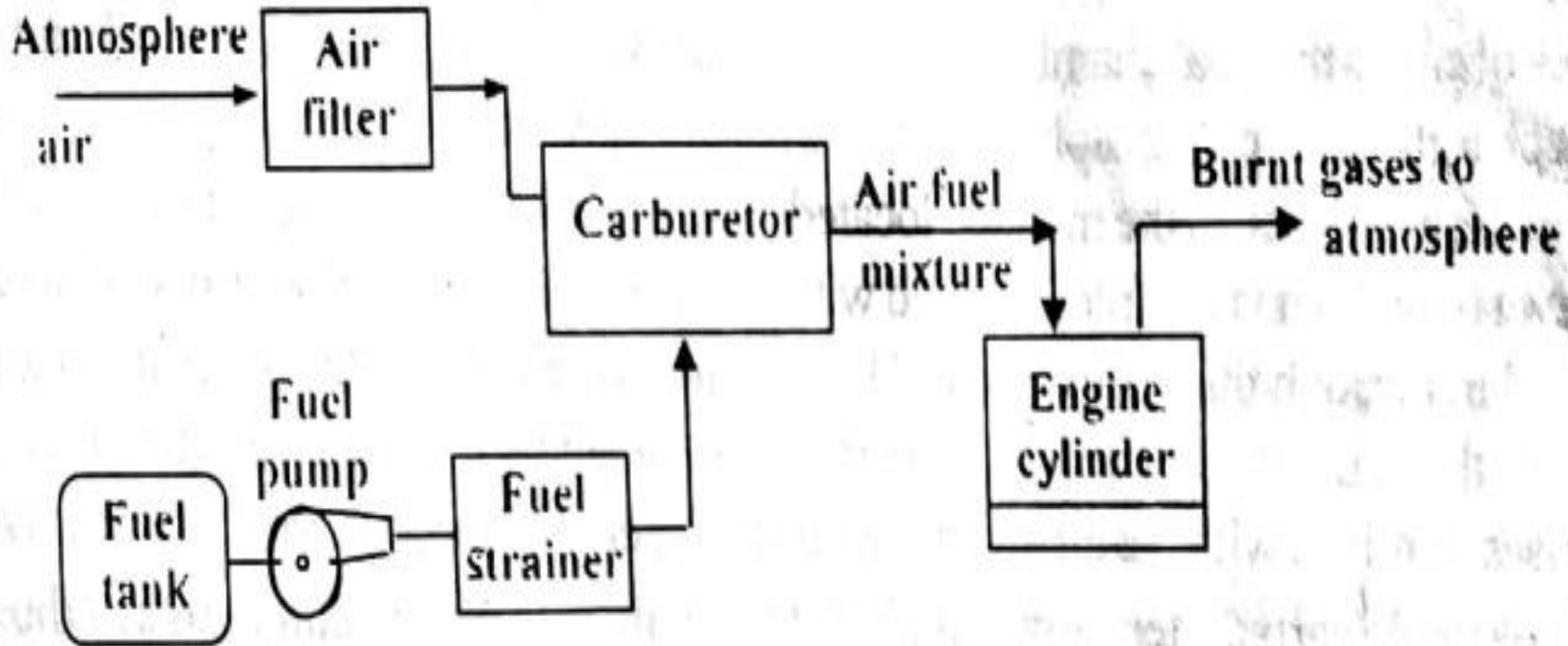
## FUEL SYSTEM FOR DIESEL ENGINES

Fuel supply system for diesel engine consists of a fuel storage tank, filter, low pressure or transfer pump, high pressure fuel pump and fuel injector. The main parts of this system are fuel pump and fuel injector.



The fuel is supplied at very high pressure from the fuel pump to the fuel injector and is injected to the engine cylinder towards the end of compression stroke.

# FUEL –AIR SYSTEM FOR CARBURETTED PETROL ENGINE

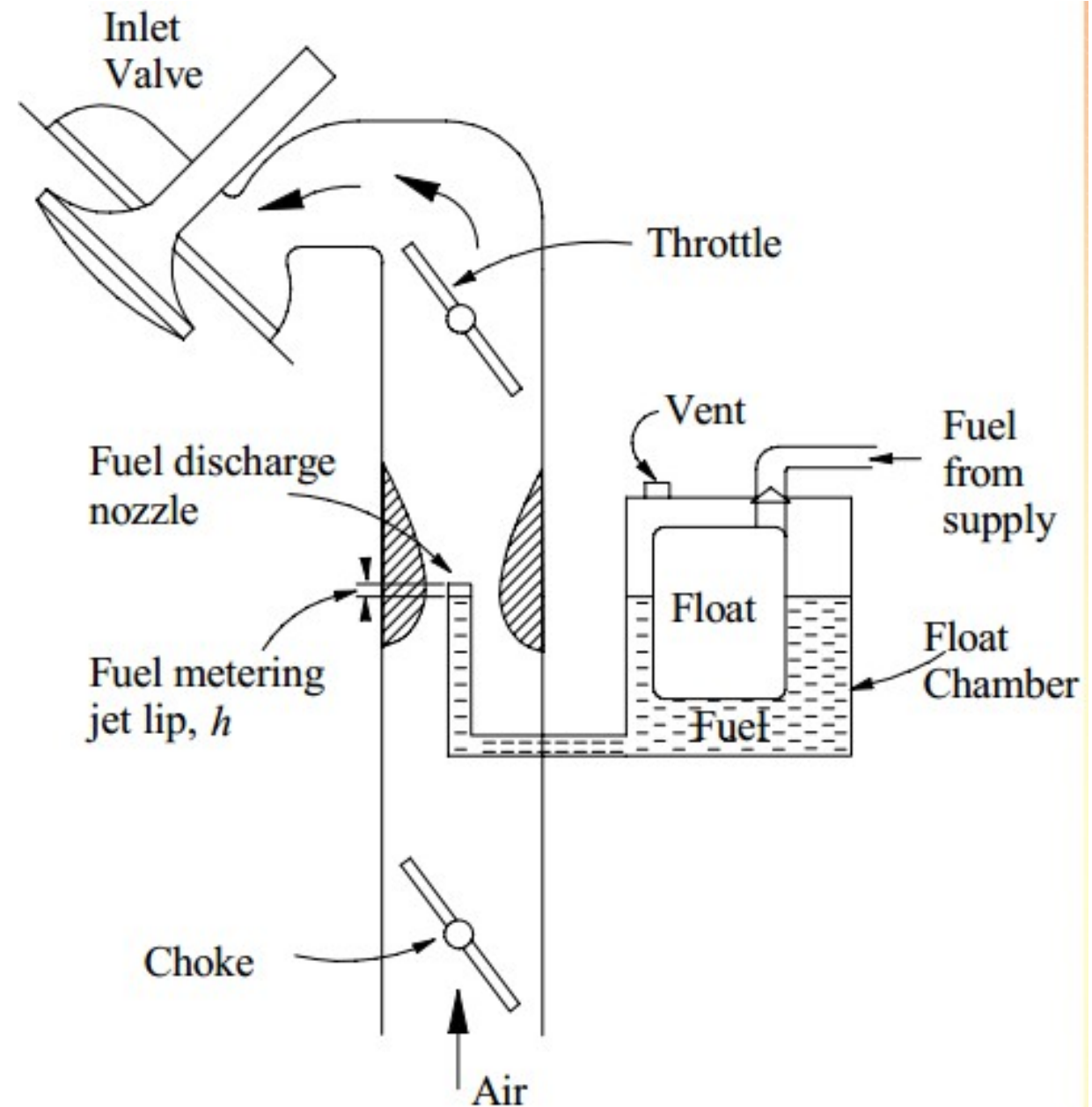


## Carburetor

- The carburetor is a device for atomising and vapourising the fuel and mixing it with the air in the varying proportions to suit the changing operating conditions of the engine.
- The process of breaking up and mixing the fuel with the air is called carburetion

# Simple Carburetor

1. Needle valve
2. Float chamber
3. Vent
4. Venturi
5. Nozzle
6. Throttle valve



- *Needle valve* attached to the float lever serves to close or open the fuel inlet to the float chamber depending upon the requirement. Thus a constant head of fuel is maintained in the *float chamber*.
- A small *vent* in the float chamber keeps the pressure inside atmospheric.
- *Venturi* is simply a restriction in the air passage. Increase in velocity & decrease in pressure is caused at the throat of the venturi where nozzle tip is located. Due to this depression being applied at the *nozzle*, the fuel comes out and is vaporized by the incoming air stream.
- The mixture then goes through the inlet manifold, to the engine cylinder.
- The purpose of a *throttle valve* is to control the quantity of air fuel mixture.

# MULTI POINT FUEL INJECTION (MPFI)

- MPFI is used in SI engines
- MPFI system injects fuel just upstream of each cylinder's intake valve, based on commands from ECU.
- In MPFI system, each cylinder has one injector placed on the side of the intake manifold near the intake valve to supply fuel in the cylinders
- Uniform air-fuel mixture will be supplied to each cylinder

## Advantages

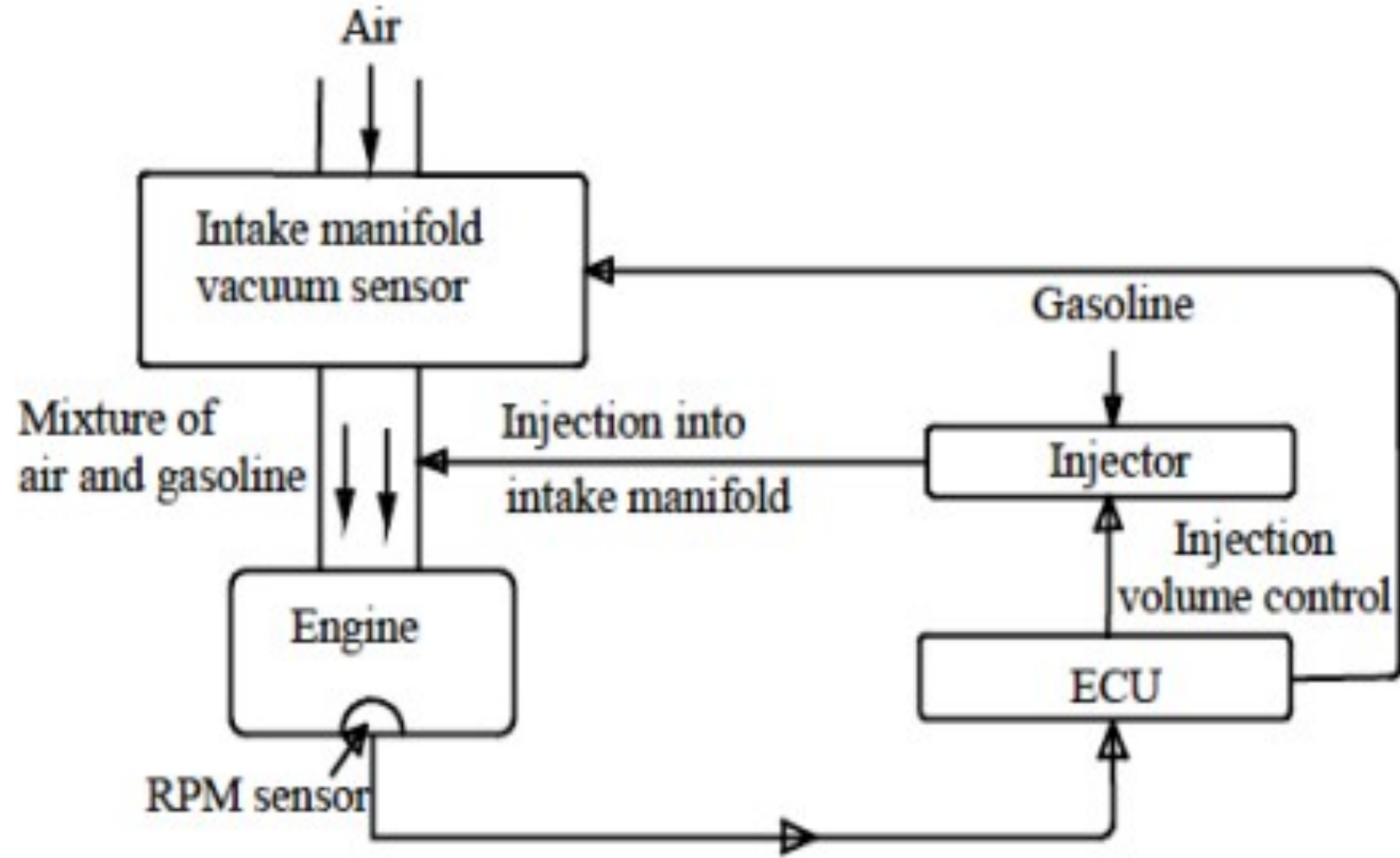
- Better fuel economy and lower emissions
- Better performance and less vibrations

## Disadvantages

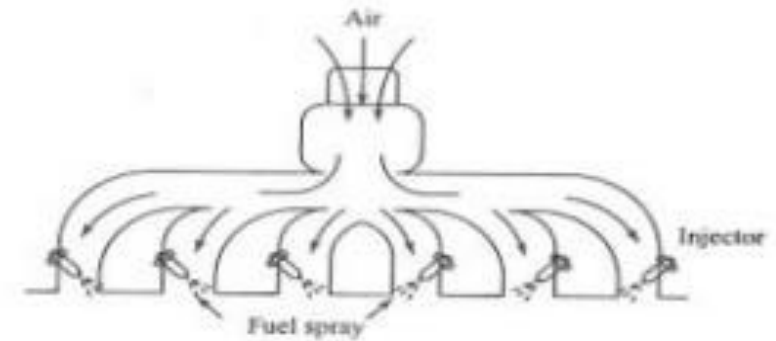
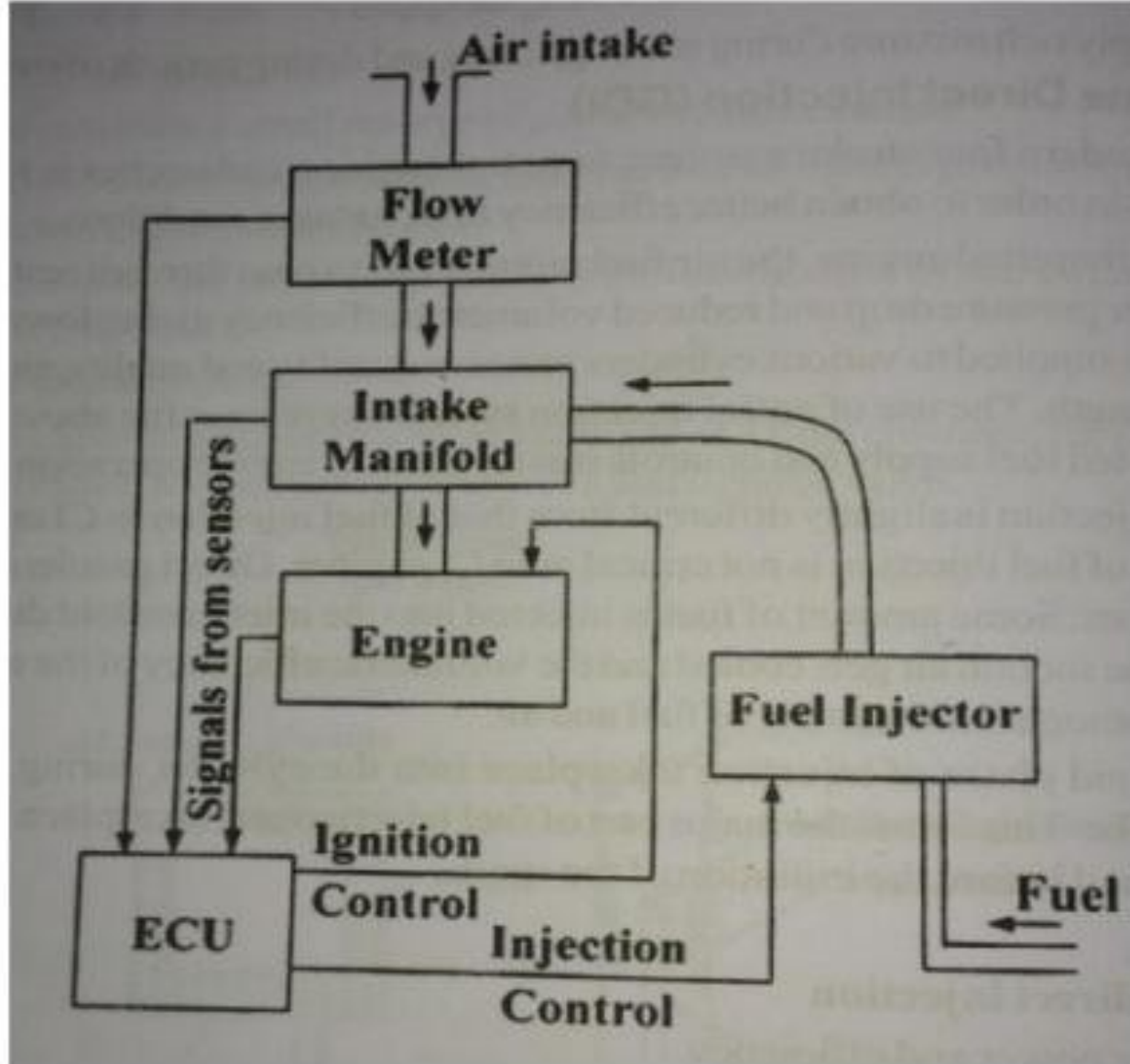
- Expensive setup
- System is complex



# MULTI POINT FUEL INJECTION(MPFI)



# MULTI POINT FUEL INJECTION (MPI)





# Methods of Engine Cooling

- Air cooling
- Liquid cooling

# Air cooling

- In this system, the heat is dissipated directly to the atmospheric air by conduction through the cylinder walls.
- In order to increase the rate of cooling, the outer surface area of the cylinder and cylinder head is increased by providing radiating fins.
- In bigger units, fans are provided to circulate the air around the cylinder walls and cylinder head.

# Air cooling

- Cooling fins
- Fan cooling

**The amount of heat dissipated depends on**

- Surface area of metal in contact with air
- Rate of air flow
- Temperature difference between the heated surface and the air
- Conductivity of the metal.

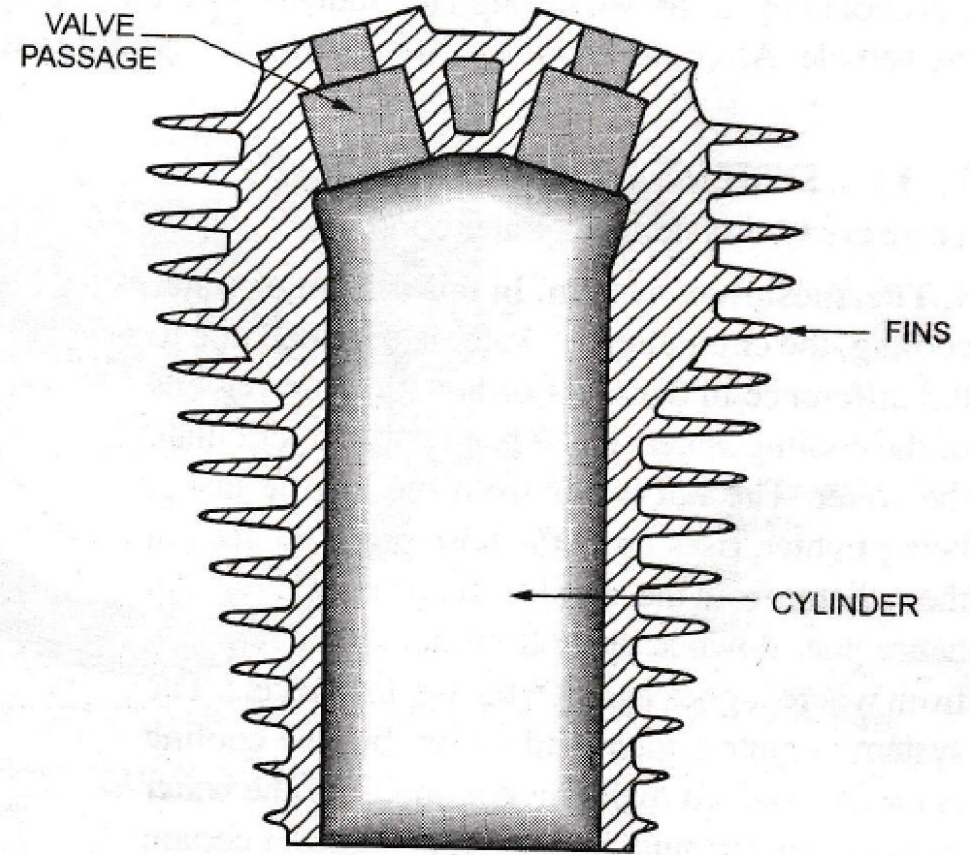
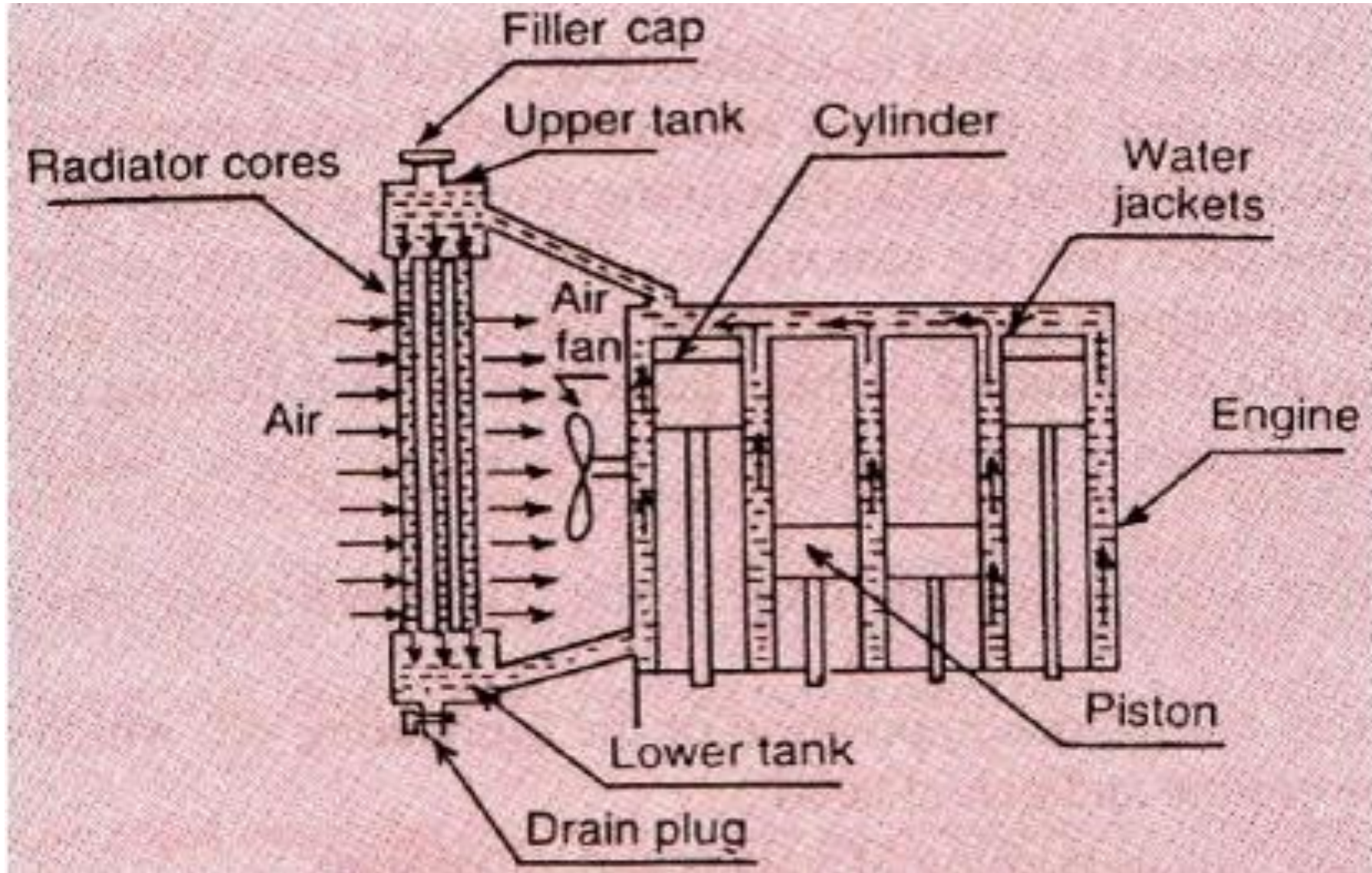


Fig. 19.1. Cylinders with fins.

## Water cooling

- In this system, the water is circulated through water jackets around each of the combustion chambers, cylinders, valve seats etc.
- The water is kept continuously in motion by a centrifugal water pump which is driven by a V-belt from the pulley on the engine crank shaft.
- After passing through the engine jackets in the cylinder block and heads, the water is passed through the radiator. In the radiator, the water is cooled by air drawn through the radiator by a fan.
- The water is again circulated through the engine jackets

## Water cooling





# Water cooling

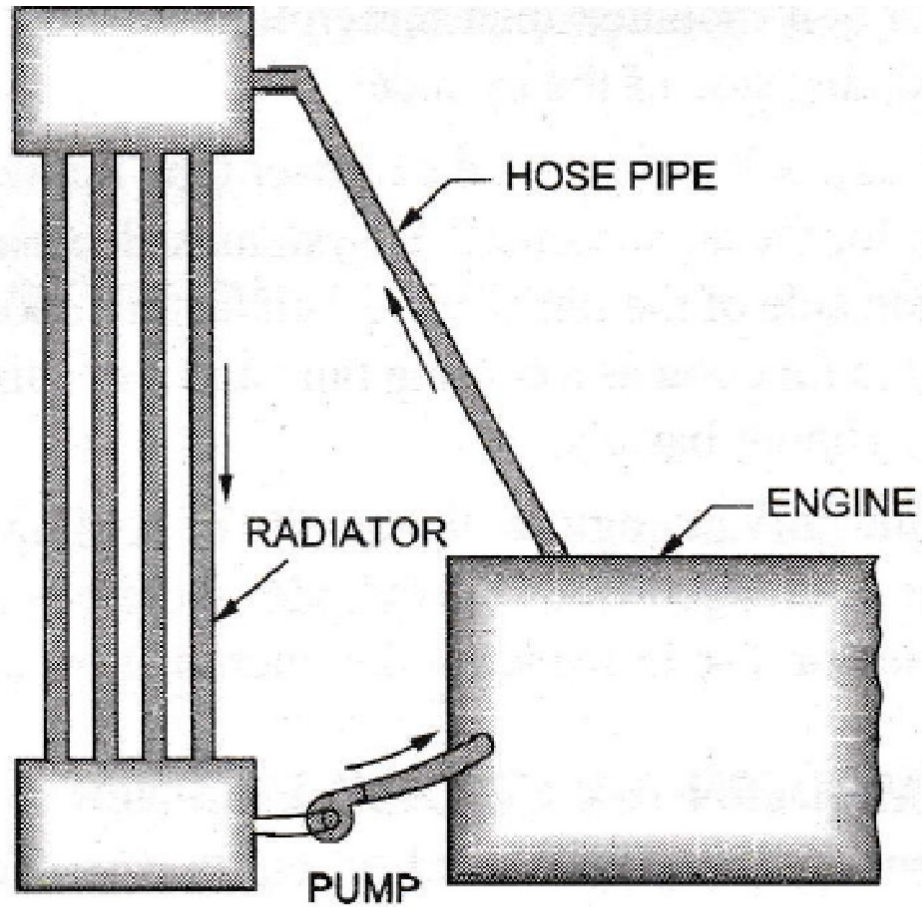


Fig. 19.3. (a) Pump circulation system of cooling.

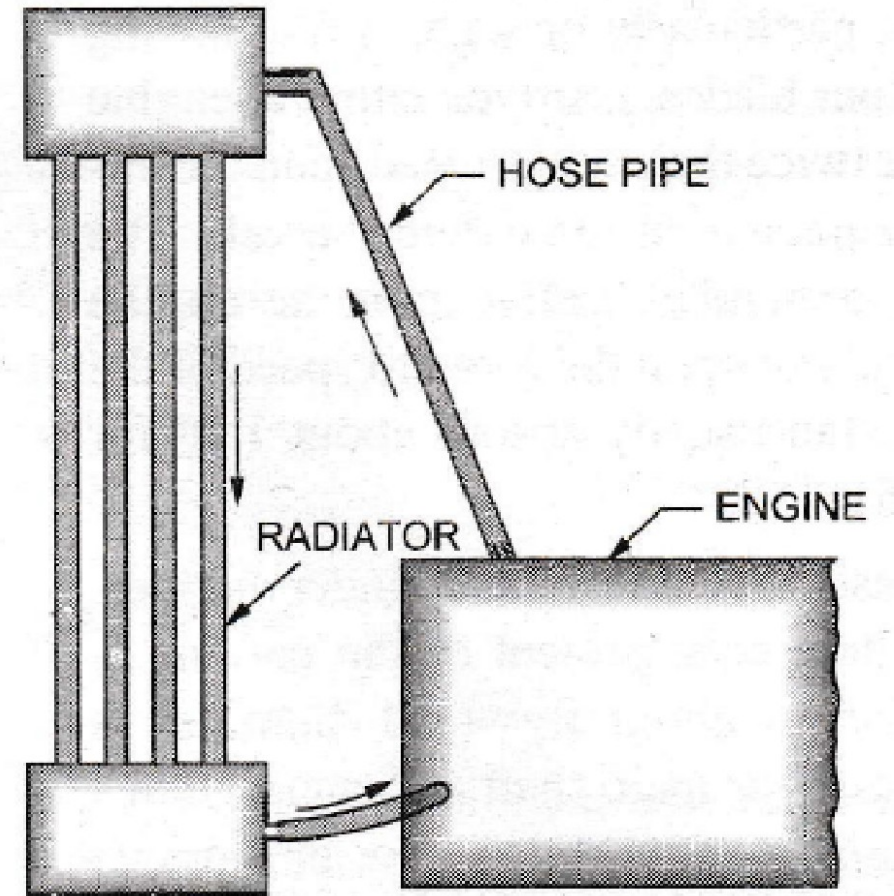


Fig.19.2. Thermosiphon system of cooling.

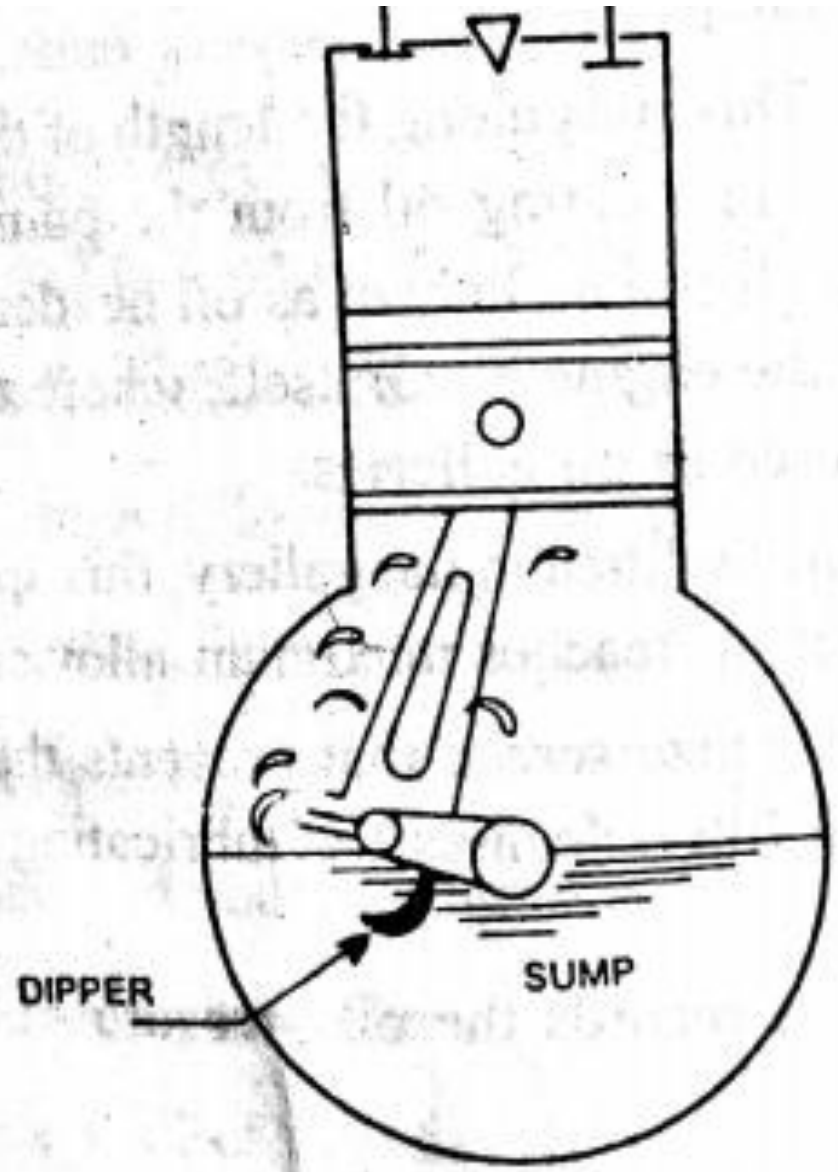


## Lubrication Systems

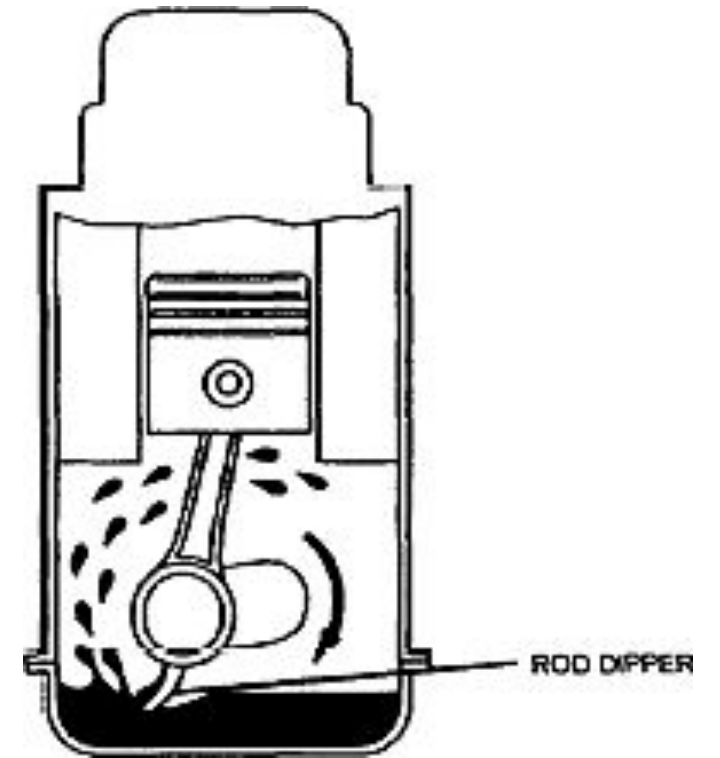
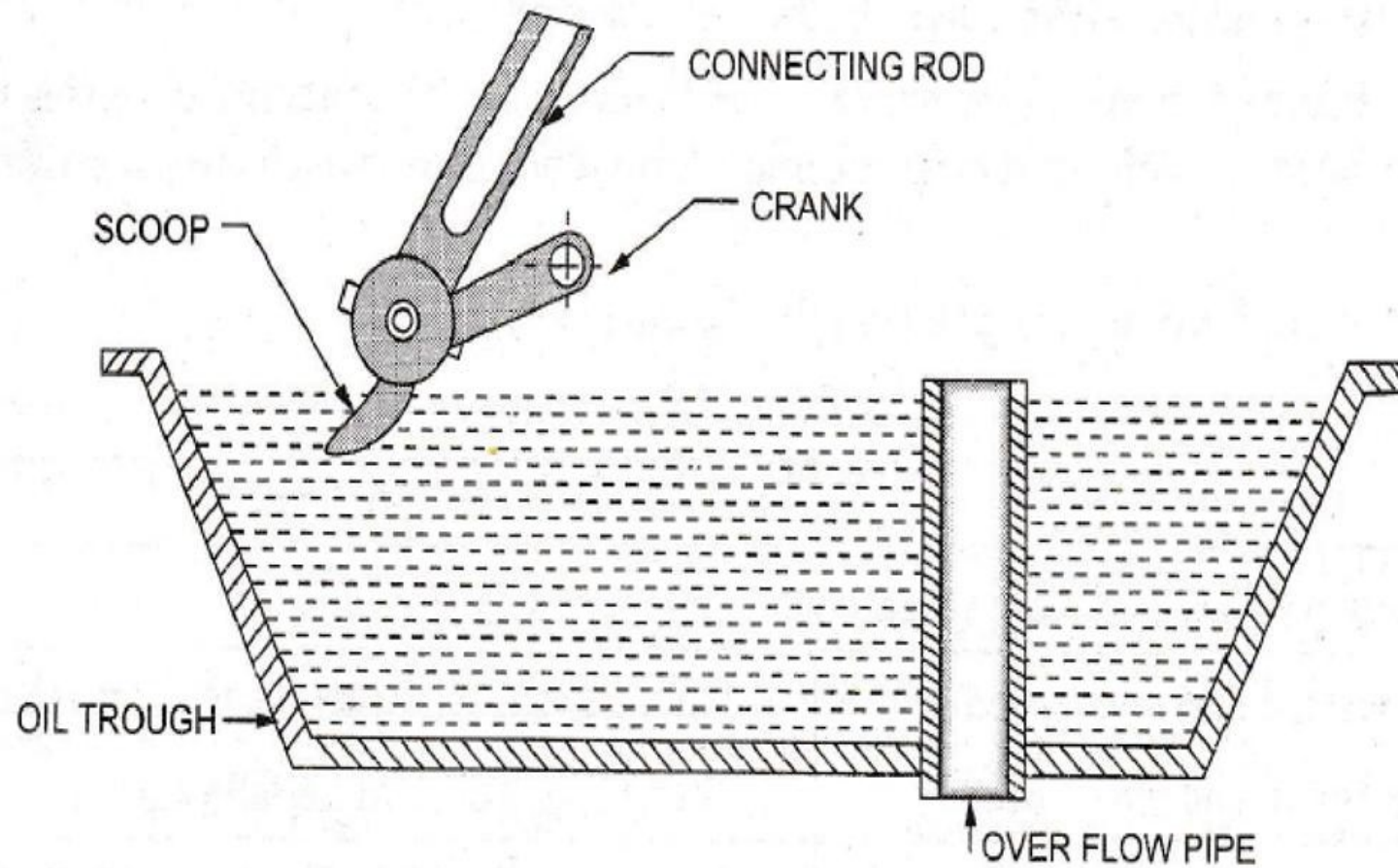
- As a matter of fact, the moving parts of an I.C. engine are likely to wear off due to continuous rubbing action of one part with another.
- In order to avoid an early wearing of the engine parts, a proper lubrication arrangement is provided in I.C. engines.
  - 1) Petroil system
  - 2) Splash system
  - 3) Pressure system

# Splash system

- This is one of the cheapest methods of the lubrication system.
- It consists of a scoop, which is fitted at the lower end of the connecting rod as shown in the diagram.
- As when the engine runs scoop splashes, the oil from oil through by centrifugal force to all engine parts.

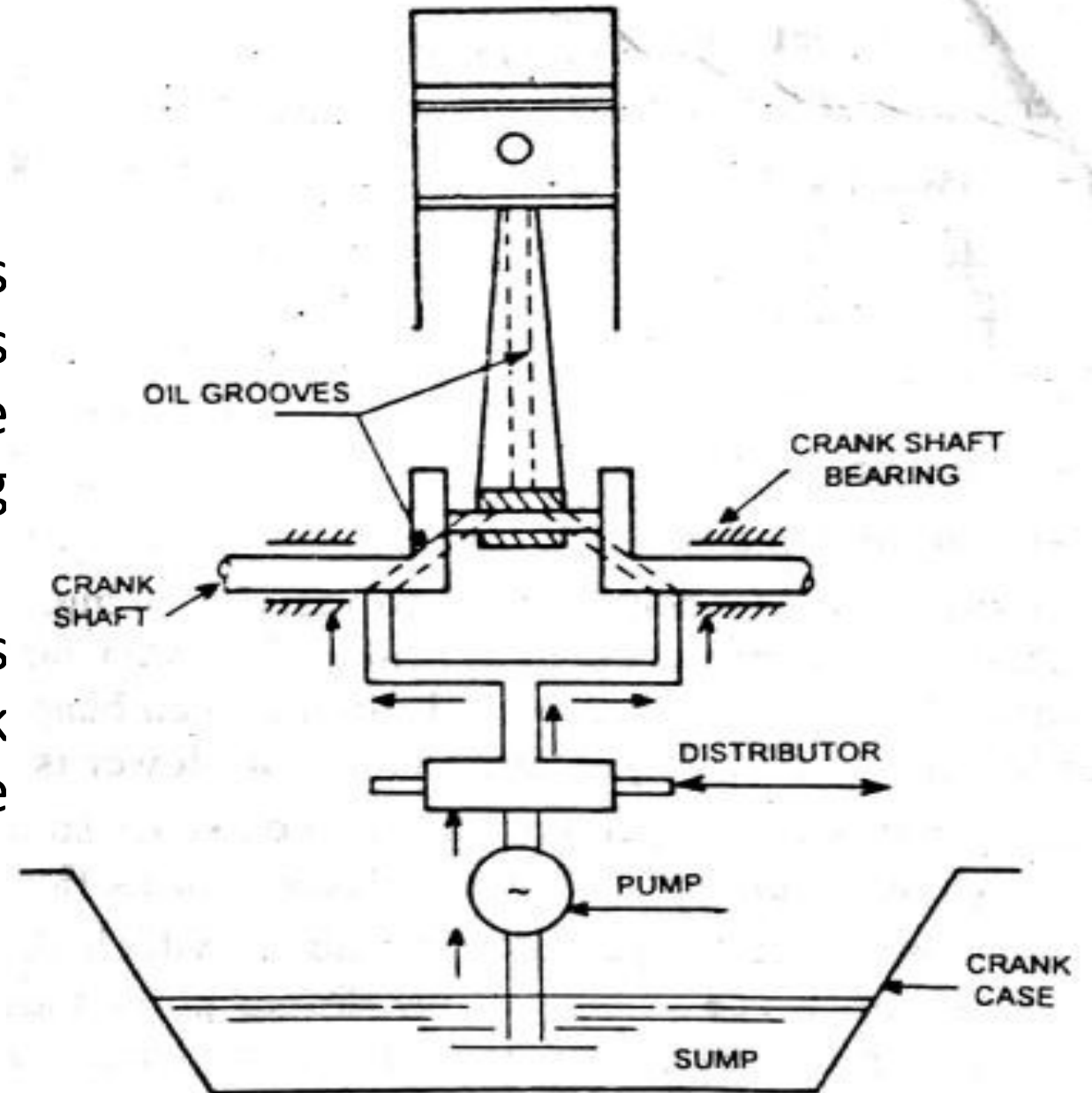


# Splash system



# Pressure system

- This method, the lubricating oil is carried in a separate tank and is pumped at a high pressure to the small gap between mating surfaces.
- It supplying oil to the various parts of engine such as crank shaft bearings, camshaft , valve mechanism etc.

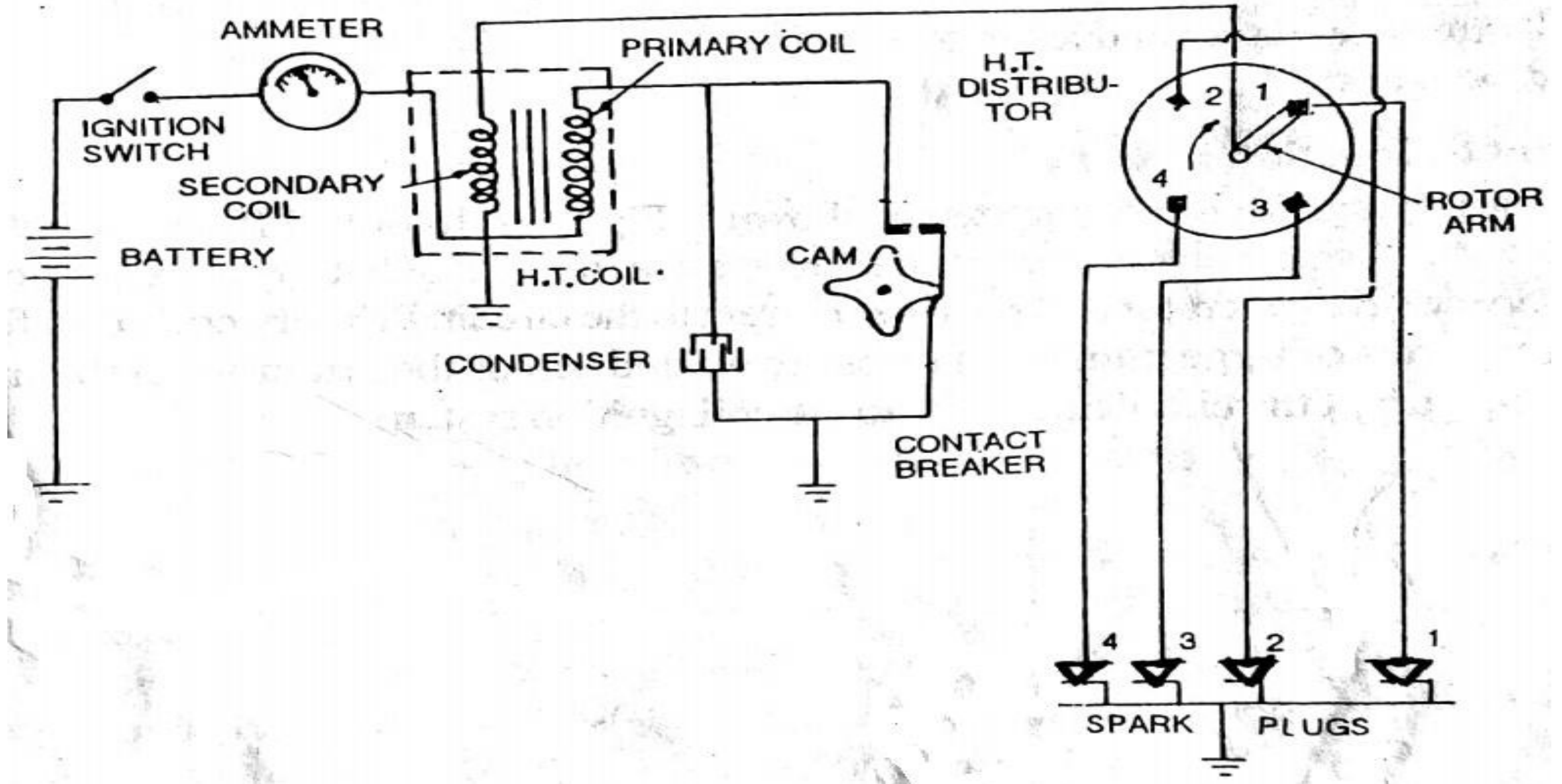


# IGNITION SYSTEMS OF PETROL ENGINES

- 1. Coil ignition system
- 2. Magneto ignition system.



# Battery Ignition System



# Magneto Ignition

